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Phase Two
Planning Phase:

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1. Identification of Need

By Lesego Mothibi

- **Defining the business case**

  The main purpose of the project is to develop an intelligent healthcare expert system that uses artificial intelligence to assist in diagnosing patient conditions such as physical symptoms and mental health concerns, while also integrating cybersecurity tools to ensure that all sensitive health data is protected. This system will offer benefits like faster and more accurate diagnostics, reduced administrative workload, and better patient engagement through secure access to their records. It also helps to prevent unauthorized system access through biometric verification and smart scanners.

  NeuroNexus HMS aims to revolutionize this landscape by offering:
  - Real-time diagnostic support using AI algorithms trained on large medial datasets.
  - Biometric authentication (e.g. fingerprint and facial recognition) and smart scanner integration to ensure only authorized users' access confidential patient data.
  - Automated administrative features, reducing human errors and saving time.
  - Enhanced patient engagement through personalized, secure access to health record, test results and treatment plans via a patient portal.

  The return on investment (ROI) will be visible in multiple dimensions:
  - **Operational efficiency:** reduced diagnostic errors and faster treatment turnaround times.
  - **Cost savings:** decreased reliance on manual paperwork and legacy systems.
  - **Cybersecurity resilience:** minimized risk of data breaches and legal liabilities.
  - **Improved patient trust and satisfaction:** leading to better health outcomes and institutional reputation.

  These benefits support the project's value and potential return on investment through better healthcare outcomes and operational efficiency. The return on investment will be seen in reduced operational risks, better data protection, and enhanced decision-making processes by healthcare professionals.

- **Data collection**

  To justify the need for the system, data must be gathered from existing healthcare platforms that rely on manual diagnosis, unsecured data handling, or outdated record management practices. These include challenges such as delays in diagnosing conditions, lack of consistent monitoring, difficulty in data sharing, and rising digital security threats targeting patient information. This data can come from interviews, usage logs, digital incident reports, or system performance reviews. Collecting this information helps to measure the severity of current inefficiencies and estimate how much improvement the proposed system can bring.

This data can come from:

- Clinical audits showing frequent misdiagnoses or delayed results.
- Surveys/interviews with healthcare providers expressing frustration with inefficiencies or security gaps.
- Incident reports detailing unauthorized access or data loss.
- Technology assessments of current tools showing lack of scalability, interoperability or compliance with evolving standards. (POPIA)

Initial findings suggest that many healthcare facilities continue to rely on manual recordkeeping or outdated software that is prone to human error, lacks integration capabilities and cannot effectively defend against modern cyber threats. Such inefficiencies compromise not only patient care but also institutional performance. The date collected will be used to model future projections, benchmark Key Performance Indicators (KPIs) and quantify the potential improvement NeuroNexus HMS could deliver.

- **Stakeholder alignment**

  Stakeholder alignment ensures the project has the support and involvement of all relevant users, such as doctors, nurses, IT professionals, system administrators, and patients. Early discussions reveal that medical professionals need a system that is not only smart but also secure, user-friendly, and capable of evolving with medical advancements. Patients, on the other hand, seek transparency, accessibility to their information, and assurance that their data remains confidential. Ensuring that these stakeholder needs are recognized and integrated into the project helps to build trust and increases the likelihood of successful system adoption.

  Stakeholders for NeuroNexus HMS include:

  - Clinical staff (doctors, nurses, specialists) who require intelligent support tools and seamless access to patient histories and lab data.
  - IT and cybersecurity personnel, who must maintain system integrity, handle technical issues and ensure compliance with data privacy regulations.
  - Administrative users, who need automation tools for patient scheduling, billing and reporting.
  - Patients, who expect a secure, transparent and easy-to-use platform to manage their health records.

Early consultations have shown strong interest and shared vision across all groups. For example, doctors value the diagnostic AI and predictive analytics, while patients emphasize the importance of data privacy and mobile accessibility. Ensuring that these perspectives are built into the system's development roadmap fosters greater trust, smooth implementation and long-term adoption.

2. Preliminary Investigation

By Zenzo Msimanga

To assess the viability of this project, we reviewed similar existing systems, conducted on-site observations, and analyzed current workflows. We also looked at technical constraints and identified both user needs and system requirements.

- **Review of Existing Systems**

In that which health related expert systems and we have many of them out there do we see full representation of the issues at hand for deaf patients which at the same time support health care workers in real time. For instance, The CSIR (Council for Scientific and Industrial Research) partnered with the NDoH to develop this system, which allows for the registration and identity verification of patients at public health facilities but what it does not do is to include access features for people with hearing impairments: AI-assisted diagnostic tools are being used in some areas, for example, to help radiologists detect abnormalities in imaging like for TB and cancer with greater speed and precision. The NDoH has an eHealth Strategy and a mHealth (mobile health) strategy that aim to leverage technology to improve access to health services, facilitate real-time data management, and empower patients with information. Private medical schemes and healthcare providers in South Africa are also heavily investing in and using AI for personalized healthcare recommendations, predicting health risks, and optimizing treatment pathways.

Another, Prognosis: Your Diagnosis, is for medical training and simulates real cases. It is useful for health professionals, but not for real time clinical work and it has no features to support disability. On the accessibility front, an app called SignAble originating from India, leverages video calls to connect deaf users to human hand sign interpreters. However, it suffers from not having tight integration to clinical decision systems and is completely dependent on the human interpreters thereby adding to the turn-around time and availability issues.

This gap in the market shows an opportunity: no single platform currently supports both expert medical decision-making and real-time communication assistance for deaf patients. Most systems either support clinicians or support people with disabilities but not both at the same time, and not in a way that can scale across clinics.

- **Analysis of Current Processes and Workflows**

We visited a public health clinic and a private hospital to observe day-to-day operations. The focus was on patient intake, communication flows, and how healthcare workers interact with patients with hearing impairments. One key issue was evident across both facilities: communication with deaf patients is inefficient and inconsistent. Most staff fall back on writing notes or using hand

gestures. If a family member or caregiver is not present, treatment is often delayed. In cases where an interpreter is available, there is no standard protocol for involving them early in the consultation.

Paper-based patient intake forms do not capture disability status or communication needs. This forces nurses and doctors to improvise, which leads to avoidable errors. Electronic medical records (EMRs) are used, but they offer no fields or prompts related to disability support. The entire process from check-in to diagnosis runs on outdated infrastructure with no support for specialized workflows.

- **Technical Environment**

Clinics rely on desktop computers at reception and nurse stations, while some doctors use tablets for reviewing records or test results. These devices have basic internet access but lack specialized peripherals. For example, none of the clinics have video-enabled systems to support real-time sign language interpretation. There are no integrations with voice recognition, sign recognition, or text-to-speech tools.

The software environment is limited to basic EMRs that are not designed for real-time assistance or accessibility. None of the systems observed included any form of AI support, even for routine triage. There is also no automation for tasks like appointment booking or documentation of patient interaction preferences, which could help improve the care experience for disabled patients.

- **User Needs**

Healthcare workers need tools that speed up symptom analysis and help them make more accurate decisions. They want suggestions on diagnoses, relevant medical guidelines, and drug interaction alerts. At the same time, they need ways to communicate efficiently with deaf patients without relying on handwritten notes or third-party interpreters.

Deaf patients, on the other hand, need to interact directly with medical staff without delays or confusion. They prefer sign language or visual communication interfaces. They also need speech-to-text tools so they can read what the clinician is saying in real time. Privacy is another concern patients do not want to depend on third parties to communicate sensitive medical issues.

- **Requirements**

The proposed system must include a rule-based expert engine to assist healthcare workers in diagnosing and managing patient care. The engine should provide decision support based on symptoms entered by the staff and suggest diagnoses, treatment plans, or follow-up actions. It should use a combination of structured medical data and machine learning models trained on clinical guidelines and patient records.

For communication support, the system should offer a sign language recognition module that uses the device’s camera to interpret patient input in real time. Speech-to-text conversion should be

integrated, so clinicians can speak naturally while the patient reads a real-time transcript. There should also be an option for text-based chat between staff and patient, using simple, readable language.

The system will require devices with front-facing cameras, strong processing capability, and reliable internet connections. Tablets or wall-mounted touchscreens at each point of care are ideal. Data security is critical: patient records, especially those involving disabilities or communication preferences, must be encrypted and access controlled. The system should log all interactions to support auditing and future improvements.

- **Challenges**

The biggest technical challenge is accurate, real-time sign language recognition. Sign language is not universal there are regional variations, and signs depend heavily on context. Training the system to understand local variants will take time and data. Integrating with existing EMRs is another challenge, especially in clinics with older or proprietary systems. This integration must be smooth to avoid double entry or data loss.

Staff will need training on how to use the system effectively. Some workers have never used assistive tech before and may resist the change. Clinics in rural areas may also face power and internet disruptions, so offline functionality will be necessary. Finally, there is the cost of purchasing and maintaining hardware that meets these technical needs.

- **Next Steps**

Build a prototype with core features: symptom input, decision support, and basic sign language or speech-to-text functionality. Test it in both public and private clinics with actual staff and deaf volunteers. Use feedback from these tests to improve the system before a broader rollout. Start with features that are easy to adopt such as real-time transcription before moving to more complex components like sign recognition.

3. Feasibility Study (Technical, Operational, Economical)

By Basani Palesa Mongwe

Feasibility studies play an important role in planning stages of any project to help us figure out if the project is worth pursuing. For a hospital, the stakes are extremely high because the outcomes have a direct impact on the health and well-being of the community. A good feasibility study requires rigorous analysis, strategic planning, and stakeholder participation to examine the project's viability from several perspectives.

Hospitals and other healthcare institutions are complicated systems that require tremendous investment, infrastructure, and experienced personnel. Thus, the feasibility study must include both broad objectives and precise specifics. It is critical to assess the hospital's potential influence on the community, alignment with public health priorities, and feasibility of implementation. The three common angles or factors to be looked at are Technical, Operational and Economical feasibility.

- **Technical Feasibility**
  The technical feasibility of a project refers to the assessment of whether a proposed project, in this case a healthcare management system can be successfully developed using the current state of technology. The factors to be considered and evaluated are, understanding and expertise needed to operate on technology, the availability of infrastructure to support the project's technical needs as well as the identification of any risks and constraints. A technological feasibility study helps us to:
  - Identify all the required technology, determine problems ahead and measure technical readiness of our organization (Hospital and other healthcare facilities) to implement and use a new system.
  - Hospitals deal with a lot of patients' information and data; a technical study allows us to implement data security measures and encryption to ensure that data is protected.
  - Ensure that all integrated hospital equipment and software can work effectively and smoothly across various departments within the hospital.

- **Economic Feasibility**

  This is where we conduct a study of the available resources, finances and the maintainability of our system, one of the common factors to be considered here is the measure of the return on investment, which is a detailed analysis of the efficiency of the system, its ability to reduce medical errors while improving patient satisfaction and efficiency. The overall costs and considerations are:

  - All the initial development processes and tools, software, infrastructure, hardware and staff.
  - Ongoing development costs, maintenance technical support and system upgrades.
  - Training the staff and licensing fees for software and medical devices.

- **Operational Feasibility**

  The operational feasibility is a measure of the functionality of the healthcare management system within the hospital's day-to-day operations, operational feasibility studies are put in place to implement all the stakeholder, management and end-user task, evaluate the management's resistance, effect on the environment or community and monitoring the system's process change, to fit all these requirements and ensure that all associated departments are comfortable with the system's functionality, the system should:

  - It should be easy to learn, to accommodate the hospital's staff (nurses and doctors)
  - The management system should be aligned with broad organizational goals, such as improving patient medical care and meeting the government's compliance standards.
  - The healthcare management system should have enough internal support and fit into existing workflows within the hospital without causing disruptions.

4. Project Planning

By Nokuthula Lengwate

- **Project Initiation**
  - **Stakeholder Identification:**
    Hospital administrators, medical staff, IT personnel, and patients.
  - **Project Charter:**
    A foundational document that outlines the project's purpose, goals, deliverables, timeline, and budget. It also officially appoints the project manager and team, setting clear authority and accountability.

- **Requirements Gathering and Analysis**
  - **Functional Requirements:**
    Managing patient records, medicine stock, staff schedules, AI diagnosis, and secure logins for different user types.
  - **Non-Functional Requirements:**
    Strong security, scalability to handle growth, user-friendliness, and reliable uptime.
  - **Stakeholder Interviews & Workshops:**
    Regular sessions with users and stakeholders to clarify needs, gather feedback, and create user stories that guide system features and design.

- **System Design**
  - **Architectural Design:**
    Plan the technical structure, ensuring the system is modular (easy to update) and service-oriented (components can work independently). Design the database and integration points for AI and biometrics.
  - **User Interface Design:**
    Create visual layouts (wireframes/prototypes) for each user portal, ensuring intuitive navigation and accessibility for all user groups.
  - **Security Design:**
    Develop robust security measures, such as role-based access (users only see what they need) and workflows for biometric authentication to protect sensitive information.

- **Technology Selection**
  The best tools and platforms for the project:
  - Programming languages and frameworks for building the system.
  - Database solutions for storing and retrieving data.
  - AI/ML frameworks for diagnostic support.
  - Biometric APIs for secure authentication.

- Cloud services for hosting and scalability.

- **Development Phase**
  - **Module Development:**
    Build each functional area (patient management, inventory, scheduling, AI, security) as separate modules for easier testing and maintenance.
  - **Integration:**
    Ensure all modules work together seamlessly, with smooth data sharing and process flows, including AI and biometric systems.
  - **Testing:**
    Thoroughly check each part (unit testing), how they work together (integration testing), the whole system (system testing), and security. Involve real users for final acceptance testing.

- **Deployment**
  - **Pilot Deployment:**
    Launch the system in a limited setting (e.g., one department) to identify issues and gather feedback before full rollout.
  - **Full Rollout:**
    Expand the system to the entire hospital, providing training and support to ensure smooth adoption and minimize disruption.

- **Maintenance and Support**

  Ongoing activities to keep the system running smoothly:

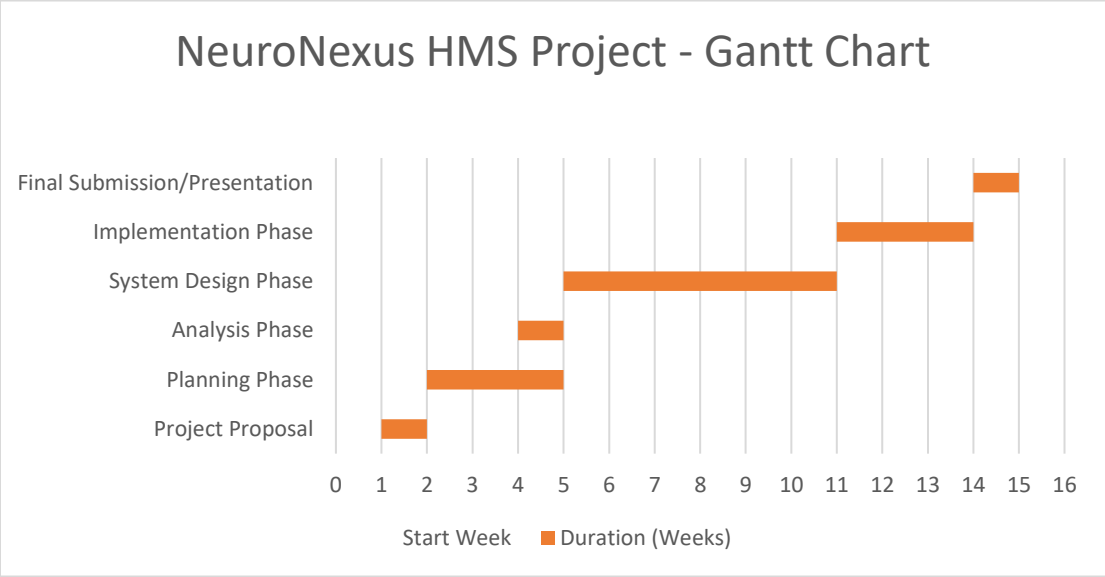
  - Provide technical support for users.
  - Regularly update software and security features.
  - Monitor performance and address any issues or user feedback promptly.

- **Project Timeline (Sample Gantt Chart)**
  - A visual schedule showing how long each phase will take, helping to track progress and ensure timely completion.

5. Project Scheduling (PERT Chart and Gantt chart)

By Zenzo Msimanga

Gantt Chart for the NeuroNexus Hospital Management System

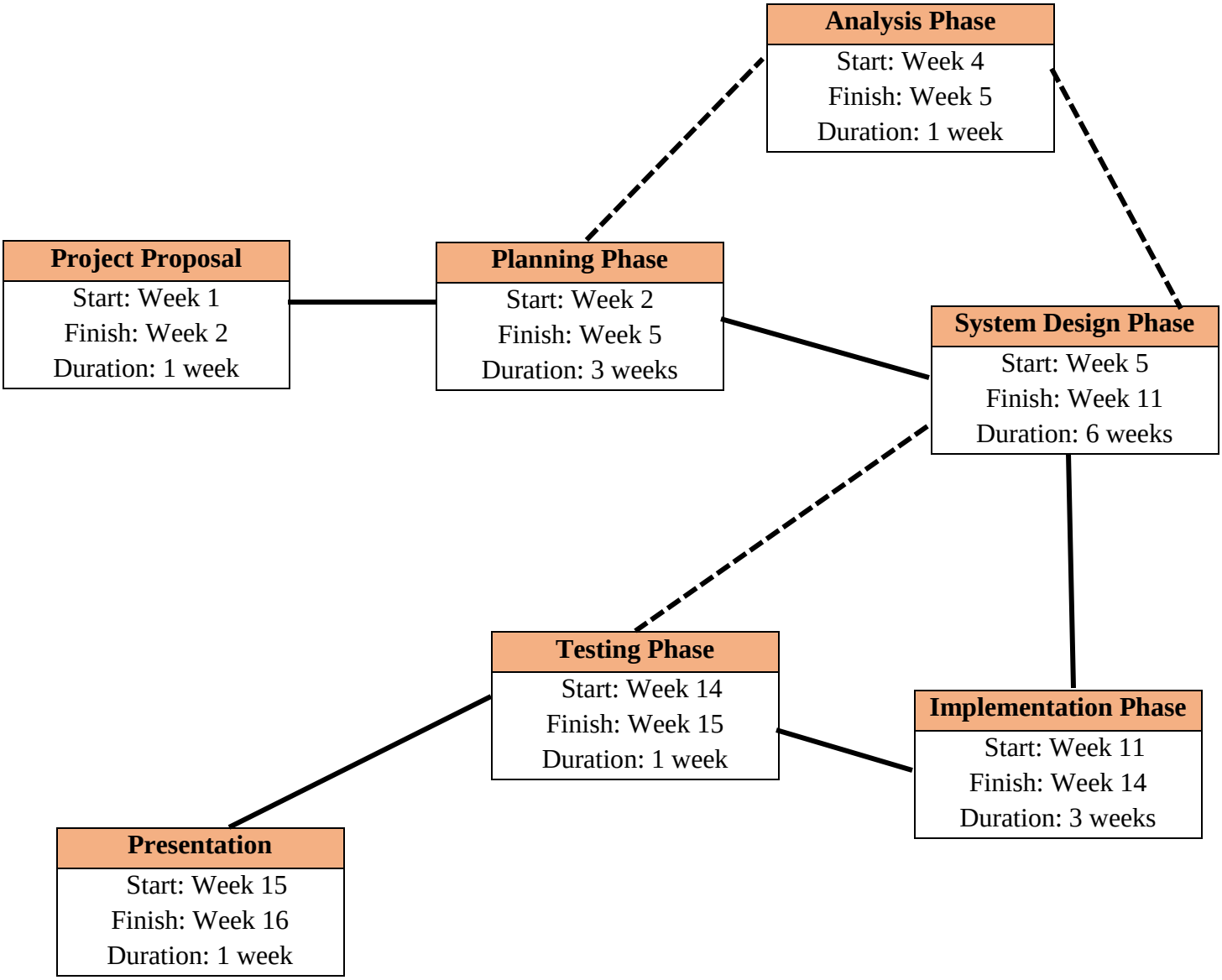


Gantt Chart representing NeuroNexus Hospital Management System Project Scheduling

Project Phase	Start Week	Duration (Weeks)
Project Proposal	1	1
Planning Phase	2	3
Analysis Phase	4	1
System Design Phase	5	6
Implementation Phase	11	3
Final Submission/Presentation	14	1

Table: Information for the Gantt Chart for the NeuroNexus HMS

PERT Chart for the NeuroNexus Hospital Management System



6. Software Requirement Specification (SRS)

By Zanele Mbatha

- **Introduction**
- Purpose

  This SRS defines the requirements for the NeuroNexus Hospital Management System (HMS), a secure, integrated digital platform for managing hospital operations, patient records, inventory, scheduling, and communication. The system aims to improve efficiency, accuracy, security, and patient care by leveraging AI-powered diagnostics and advanced access controls.

- Scope

  The NeuroNexus HMS will provide:
  - Centralized management for all hospital operations
  - Secure portals for doctors, nurses, administrative staff, pharmacists, and patients
  - AI-based expert system for clinical decision support
  - Biometric authentication (facial recognition, fingerprint scanning)
  - Role-based access control
  - Real-time data access and communication
  - Compliance with healthcare data privacy regulations (e.g. POPIA)

- Definitions, Acronyms, and Abbreviations

- HMS: Hospital Management System
- AI: Artificial Intelligence
- POPIA: Protection of Personal Information Act

- **Overall Description**
- Product Perspective

  The HMS will replace fragmented manual and digital systems with a unified platform. It will be modular and scalable, allowing for future feature expansion without disrupting existing operations.

- Product Functions
  - Secure login and authentication for all user types
  - Patient record management (creation, update, retrieval)
  - Medicine inventory management
  - Scheduling and calendar management
  - AI-powered diagnostic support for doctors
  - Communication tools (messaging, notifications)
  - Patient portal for personal health information access

- Biometric and role-based access controls
- Audit logs and activity monitoring

- User Classes and Characteristics
  - Doctors: Full access to patient records, diagnostics, and scheduling
  - Nurses: Access to assigned patient records, scheduling, and inventory
  - Administrative Staff: Access to scheduling, admissions, discharges, and billing
  - Pharmacists: Access to medicine inventory and prescription management
  - Patients: Access to own records, test results, and appointments

- Operating Environment
  - The application will be a downloadable, installable app available for Windows, macOS, and Linux desktop environments, as well as Android and iOS mobile devices.
  - Secure network infrastructure with encrypted data transmission will be utilized for all data exchanges between clients and the central server.
  - The app will operate on hospital workstations, tablets, and smartphones.
  - Secure network infrastructure with encrypted data transmission

- Design and Implementation Constraints
  - Compliance with data privacy regulations (POPIA)
  - Integration with existing hospital hardware (biometric scanners)
  - High availability and minimal downtime
  - Modular architecture for scalability
  - The application must be compatible with major desktop and mobile operating systems.

- User Documentation
  - Comprehensive user manuals for all user roles
  - Training materials and tutorials
  - Online help and support documentation

- **System Features and Requirements**
- Functional Requirements

Feature	**Description**
User Authentication	Secure login with password and biometric options (facial/fingerprint)
Role-Based Access Control	Permissions based on user roles (doctor, nurse, admin, pharmacist, patient)
Patient Record Management	CRUD (Create, Read, Update, Delete) operations on patient data

Medicine Inventory Management	Track, update, and manage medicine stocks, alerts for low inventory
Scheduling	Manage appointments, staff shifts, and resource allocation
AI-Powered Diagnosis	Analyze patient data to suggest diagnoses and treatment options for doctors
Patient Portal	Patients can view their records, test results, and appointments securely
Communication Tools	Internal messaging, notifications, and alerts for staff
Accessibility Features	Support for deaf/hard-of-hearing patients (e.g., text communication, visual alerts)
Audit Trails and Monitoring	Log user activities, monitor access, and detect suspicious behavior
Data Encryption	Encrypt sensitive data at rest and in transit
Compliance and Reporting	Generate reports for compliance with healthcare regulations

- Non-Functional Requirements

Category	**Requirement**
Performance	Real-time data updates, minimal latency for critical operations
Reliability	99.9% uptime, robust backup and recovery mechanisms
Scalability	Support for increasing users, data, and new modules
Security	End-to-end encryption, biometric authentication, role-based access, compliance with POPIA

Usability	Intuitive UI for all user types, accessible design
Maintainability	Modular codebase, well-documented APIs, easy to update and extend
Portability	The application must be installable and fully functional on Windows, macOS, Linux, Android, and iOS devices.

- **External Interface Requirements**
- User Interfaces
  - Responsive web portal for all user roles
  - Dashboard with role-specific widgets and navigation
  - Biometric authentication interface
  - Accessible design for users with disabilities

- Hardware Interfaces
  - Integration with biometric scanners (facial recognition, fingerprint, etc.)
  - Compatibility with hospital IT infrastructure

- Software Interfaces
  - APIs for integration with laboratory systems, billing, and third-party healthcare applications

- Communication Interfaces
  - Secure HTTPS protocol for all data transmission
  - Internal messaging and notification system

- **System Architecture**
  - Modular Design: Separate modules for patient management, inventory, scheduling, AI diagnostics, and security
  - Centralized Database: Unified, encrypted storage for all hospital data
  - AI Engine: Machine learning models for diagnosis, continuously updated with new data
  - Access Control Layer: Manages authentication and authorization
  - Audit and Monitoring: Real-time logging and anomaly detection

- **Other Requirements**
  - Training: Onboarding sessions and ongoing training for all user groups
  - Support: 24/7 technical support and maintenance
  - Documentation: Up-to-date user and system documentation
  - Testing: Comprehensive testing (unit, integration, user acceptance) before deployment

- **Appendices**
  - Glossary of Terms
  - Regulatory Compliance Checklist
  - Sample User Flows and Screenshots (to be developed in design phase)

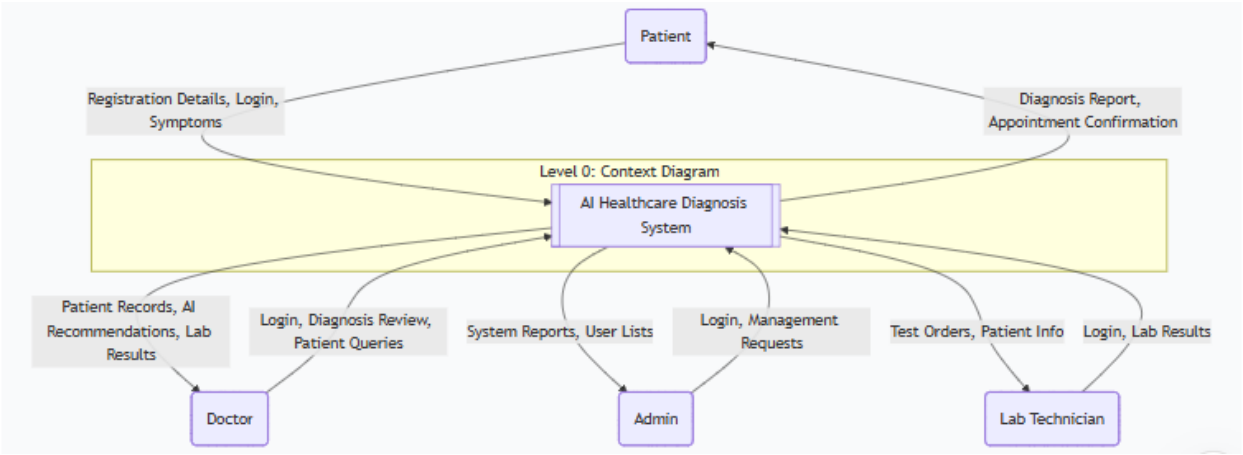
- **Conclusion**

The NeuroNexus Hospital Management System (HMS) is designed to modernize and streamline hospital operations through a secure, feature-rich, and user-friendly downloadable application. By consolidating patient management, inventory, scheduling, and communication into a single platform, the HMS will enhance efficiency, accuracy, and patient care. The integration of AI-powered diagnostics, robust security measures, and compliance with healthcare regulations ensures that the system meets the highest standards of safety and reliability. With its modular architecture and cross-platform compatibility, the NeuroNexus HMS is well-positioned to adapt to the evolving needs of healthcare institutions, providing a scalable foundation for future growth and innovation.

7. Data Models

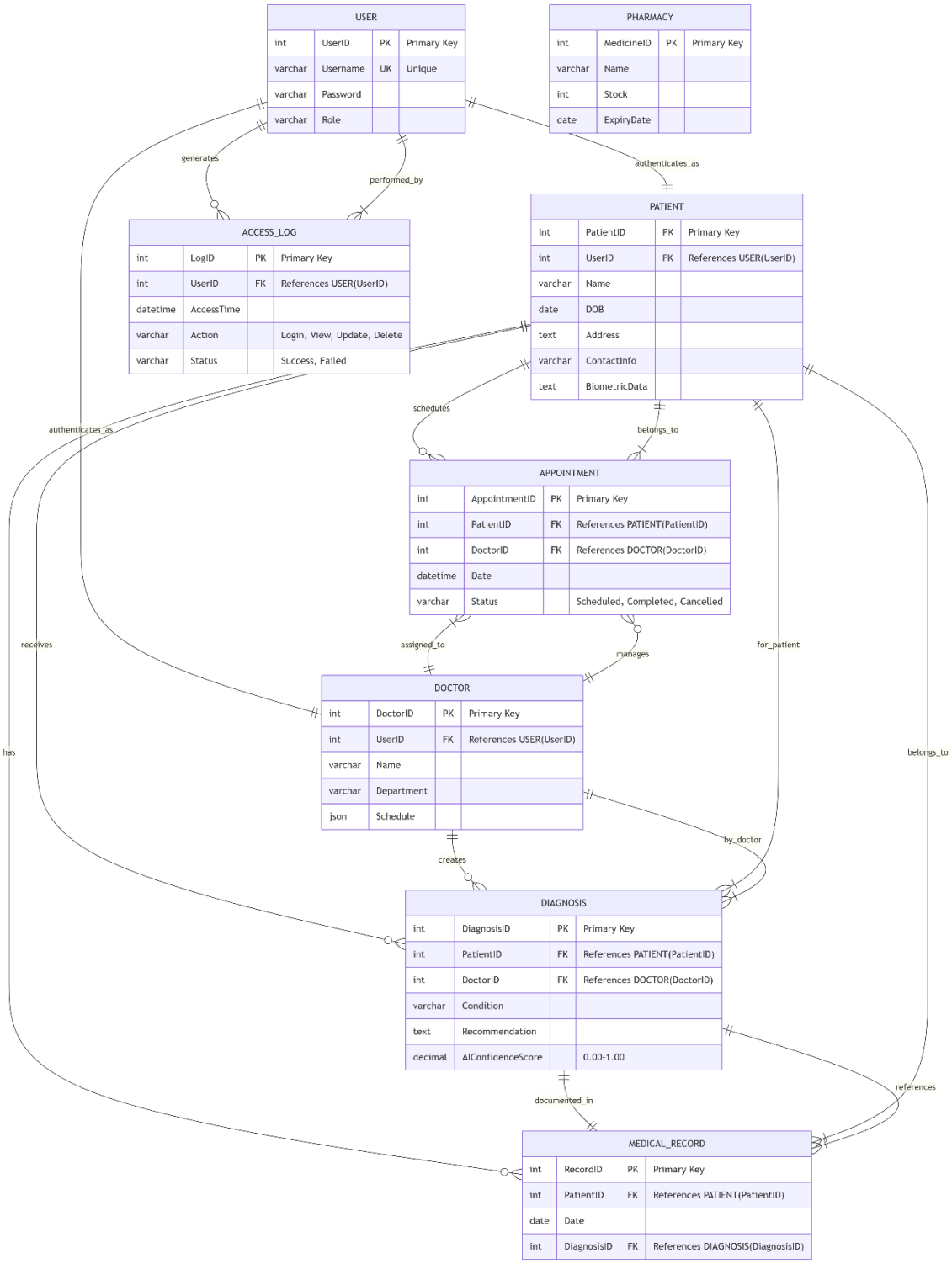
By MandlaJr Nhlapo

- **Data Flow Diagrams (DFD)**
  - **Purpose**
    To show how data moves through the HMS – from the inputs (login, diagnosis requests) to processing (AI engine and authentication) and outputs (patient record updates, reports)
  - **Components**
    External entities such as Patient, Doctor, Admin and Lab Technician
  - **Processes**
    1. Patient Registration/Login
    2. AI-Assisted Diagnosis
    3. Record Management
    4. Access Control and Authentication
  - **Data Stores**
    1. Patient Database
    2. AI Model Repository
    3. Logs and Access Records
  - **Data Flows**
    1. User credentials → Authenticate → Access granted
    2. Symptoms → AI Systems → Diagnosis → Medical record update
  - Tool
    1. Lucidchart
    2. Draw.io



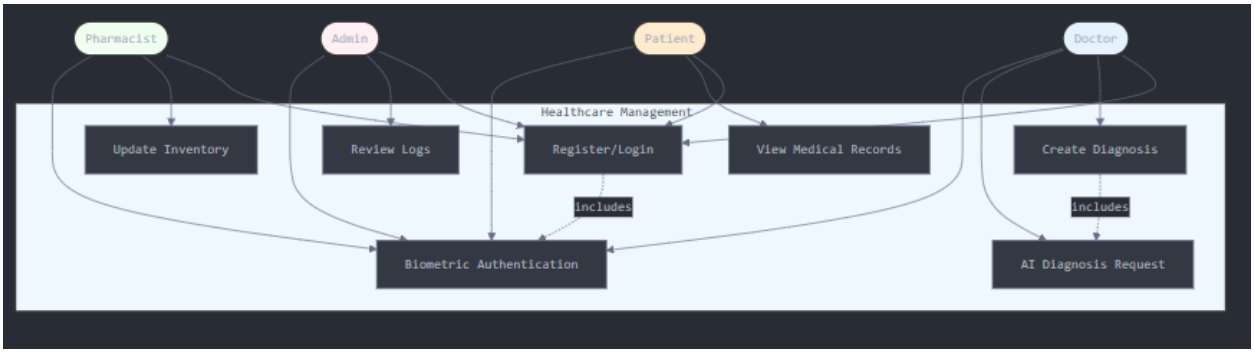
Data Flow Diagram of NeuroNexus Hospital Management System

- **Entity Relationship Diagram (ERD)**
  - **Purpose**
    To design the database structure showing relationships between hospitals entities.
  - **Key Entities and Attributes**
    1. Patient (PatientID, Name, DOB, Address, ContactInfo, BiometricData)
    2. Doctor (DoctorID, Name, Department, Schedule)
    3. Appointment (AppointmentID, PatientID, DoctorID, Date, Status)
    4. Diagnosis (DiagnosisID, PatientID, DoctorID, Condition, Recommendation, AIConfidenceScore)
    5. Medical_Record (RecordID, PatientID, Date, DiagnosisID)
    6. User (UserID, Username, Password, Role)
    7. Access_Log (LogID, UserID, AccessTime, Action, Status)
    8. Pharmacy (MedicineID, Name, Stock, ExpiryDate)
  - **Relationships**
    1. Patient has many Appointments
    2. Doctor manages many Diagnoses
    3. Patient has one or many Medical_Records
    4. Each user generates may Access_Logs



Entity-Relationship Diagram for NeuroNexus Hospital Management System

- **Use-Case Diagram**
  - **Purpose**
    Shows how users interact with the system.
  - **Actors**
    1. Patient
    2. Doctor
    3. Admin
    4. Pharmacist
  - **Use-cases**
    1. Register/Login
    2. View Medical Records (Patient)
    3. Create Diagnosis (Doctor)
    4. Update Inventory (Pharmacist)
    5. Review Logs (Admin)
    6. Biometric Authentication
    7. AI Diagnosis Request

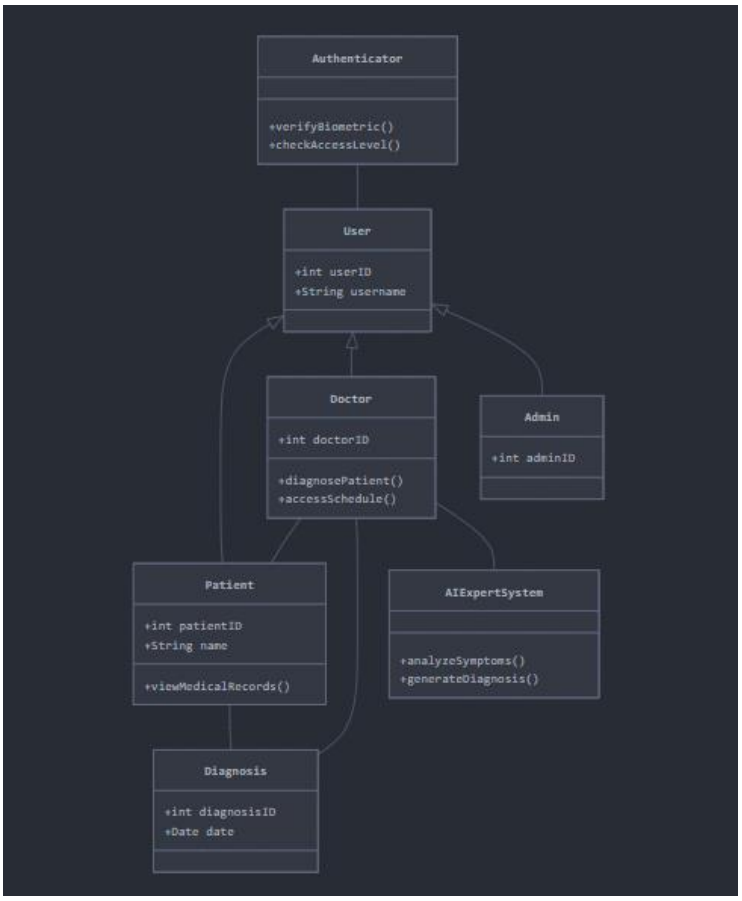


- **Class Diagram**
  - **Purpose**
    Object-oriented representation of the system.
  - **Examples of Classes:**
    1. Class: Patient
       a. patientID: int
       b. name: String
       c. viewMedicalRecords()
    2. Class: Doctor
       a. doctorID: int
       b. diagnosePatient()
       c. accessSchedule()
    3. Class: AIExpertSystem

a. analyzeSymptoms()
b. generateDiagnosis()

4. Class: Authenticator
   a. verifyBiometric()
   b. checkAccessLevel()

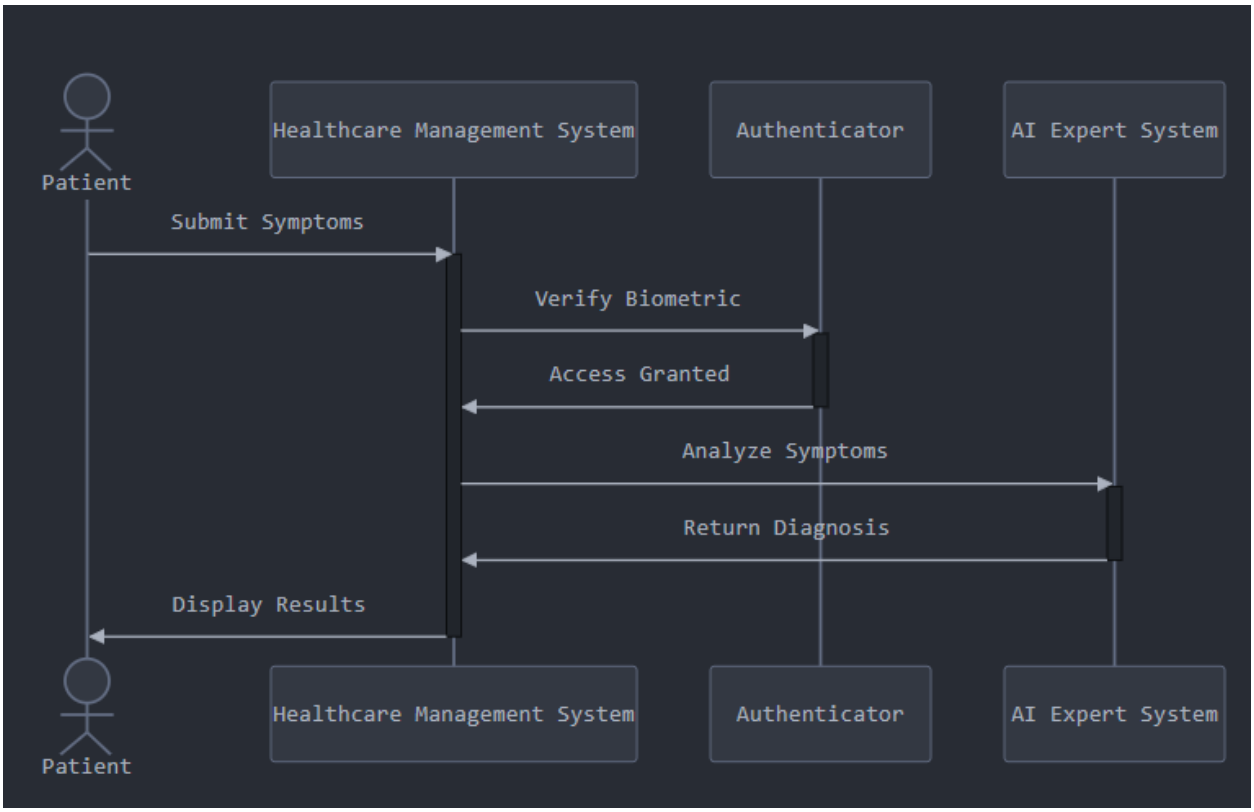
- **Relationships**
  1. Inheritance (User ← Patient, Doctor, Admin)
  2. Association (Doctor ↔ Patient via Diagnosis)



Class Diagram for the NeuroNexus Hospital Management System

- **Sequence Diagram**
  - **Purpose**
    Shows the flow of operations during AI-assisted diagnosis.
  - **Example flow:**
    1. Patient → HMS: Submit Symptoms
    2. HMS → Authenticator: Verify Biometric
    3. Authenticator → HMS: Access Granted
    4. HMS → AIExpertSystem: Analyze Symptoms
    5. AIExpertSystem → HMS: Return Diagnosis

6. HMS → Patient: Display Results



Sequence diagram for NeuroNexus Hospital Management System

- **State Diagram**
  - **Purpose**
    Models the state transitions of a diagnosis record.
  - **States:**
    New Request → Under Review → AI Processed → Doctor Validated → Closed
- **Activity Diagram**
  - **Purpose**
    Shows patient workflow from login to receiving treatment.
  - **Activities**
    Login → Authenticate Biometric → Choose Service → View Diagnosis → Book Appointment → Logout

- **Class Responsibility Collaboration (CRC) Cards**

Class	Responsibilities	Collaborators
Patient	View medical records, input symptoms, book appointments	Doctor, SystemAuth
Doctor	View patient data, issue diagnoses, prescribe treatment	Patient, AIExpertSystem
AIExpertSystem	Analyze patient input, generate diagnosis, recommend treatment	Doctor, MedicalDatabase
SystemAuth	Authenticate users, manage role-based access, log sessions	All Users, AccessLog

- **Tools we will use**
  - Draw.io: For DFDs, ERDs, UML Diagrams
  - StarUML or Visual Paradigm: Full UML toolset
  - PlantUML (with Markdown support): For Sequence/Class Diagrams
  - ERDPlus: Easy-to-use ER diagram creation

8. References

1. Health Data Management. (2025). Why AI must increasingly power cybersecurity in healthcare - Health Data Management. [online] Available at: https://www.healthdatamanagement.com/articles/why-ai-must-increasingly-power-cybersecurity-in-healthcare?utm_source [Accessed 11 Jun. 2025].
2. Putty, C. (2025). Safeguarding Patient Data: AI's Role in Healthcare Cybersecurity. [online] Thoughtful.ai. Available at: https://www.thoughtful.ai/blog/safeguarding-patient-data-ais-role-in-healthcare-cybersecurity?utm_source [Accessed 11 Jun. 2025].
3. Transcend Digital (2025). Healthcare Data Management Challenges and Solutions: 2024 Guide for IT Leaders - Transcend Digital. [online] Transcend Digital -. Available at: https://transcenddigital.com/blog/healthcare-data-management-challenges-and-solutions-guide-for-it-leader/?utm_source [Accessed 11 Jun. 2025].
4. Singhal, A., Neveditsin, N., Tanveer, H. and Mago, V. (2024). Toward Fairness, Accountability, Transparency, and Ethics in AI for Social Media and Health Care: Scoping Review. JMIR Medical Informatics, [online] 12(1), p.e50048. doi: https://doi.org/10.2196/50048.
5. Stoumpos, A.I., Ktsios, F. and Talias, M.A. (2023). Digital Transformation in healthcare: Technology Acceptance and Its Applications. International Journal of Environment Research and Public Health, [online] 20(4), p.3407. Available at: https://www.mdpi.com/1660-4601/20/4/3407.
6. Justice, V. (2024).Navigating the digital transformation in healthcare: challenges and solutions. [online] MedicalEconomics. Available at: https://www.medicaleconomics.com/vew/navigatig-the-digital-transformation-in-health-care-challenges-and-solutions?utm_source [Accessed 12 June. 2025]
7. HIMSS (2020). Interoperability in Healthcare. [online] Available at: https://legacy.himss.org/resources/interoperability-healthcare?utm_source [Accessed 12 June. 2025]
8. Hart, K.W., Rucker, D.W. and Jacobs, B.R. (2012). Reduction in Clinical Variance Using Targeted Design Changes in Computerized Provider Order Entry (CPOE) Order Sets. Applied Clinical Informatics, 03(01), pp.52-63. doi: https://doi.org/10.4338/aci-2011-01-ra-002.
9. HealthIT.gov(2019). Benefits of EHRs.[online] www.healthit.gov. Available at: https://www.healthit.gov/topic/health-it-and-health-information-exchange-basics/benefits-ehrs.
10. Shortliffe, E. H., & Cimino, J. J. (Eds.). (2014). *Biomedical Informatics: Computer Applications in Health Care and Biomedicine*. Springer.
11. World Health Organization. (2020). *Electronic Health Records: Manual for Developing Countries*.

12. Topol, E. (2019). *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*. Basic Books.
13. Rajkomar, A., Dean, J., & Kohane, I. (2019). "Machine Learning in Medicine." *New England Journal of Medicine*, 380(14), 1347-1358.
14. ISO/IEC 27001:2013 Information Security Management Standards.
15. U.S. Department of Health & Human Services. (2022). *Health Information Privacy (HIPAA)*.
16. Jain, A. K., Ross, A., & Nandakumar, K. (2011). *Introduction to Biometrics*. Springer.
17. Microsoft Azure Face API Documentation: https://azure.microsoft.com/en-us/services/cognitive-services/face/
18. Project Management Institute. (2017). *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*.
19. LazyMonkey, 2025. SRS for Hospital Management Systems. online Available at: https://lazymonkey.in/blog/srs-for-hospital-management-systems/ Accessed 12 June 2025.
20. Scribd, 2025. SRS For Hospital Management System. online Available at: https://www.scribd.com/document/516860633/SRS-for-hospital-management-system Accessed 12 June 2025.
21. SlideShare, 2025. Hospital Management System SRS for software. online Available at: https://www.slideshare.net/slideshow/hospital-management-system-srs-for-software/266492971 Accessed 12 June 2025
22. -Scribd, 2025. SRS FOR HOSPITAL MANAGEMENT SYSTEM - Odt. online Available at: https://www.scribd.com/document/403974628/SRS-FOR-HOSPITAL-MANAGEMENT-SYSTEM-odt Accessed 12 June 2025.
23. Okoromi, E.J., 2025. Hospital Management System (HMS). DiVA portal. online Available at: https://www.diva-portal.org/smash/get/diva2:1852404/FULLTEXT01.pdf Accessed 12 June 2025.
24. Grafiati, 2025. Bibliographies: 'Hospital Management Systems'. online Available at: https://www.grafiati.com/en/literature-selections/hospital-management-systems/ Accessed 12 June 2025.
25. MSCW, 2025. Hospital Management System Software Engineering Project Report. online Available at: https://mscw.ac.in/NAAC/Criteria1/Samples-of-ProjectWork_Fieldwork/Computer_Science/software_Engineering/Software%20Engineering/Hospital%20Managment%20Project%20SE-converted.pdf Accessed 12 June 2025.
26. Anon., 2021. *Health Level 7 International.* [Online] Available at: https://www/hl7.org/implement/standards/ [Accessed 12 June 2025].
27. Anon., 2025. *Aninver* [Online] Available at: https://aninver.com/blog/feasibility-study-of-healthcare-centers-and-hospitals-key-considerations [Accessed 10 June 2025].
28. Craig Fukushima, N., 2025. *foxgrp.com* [Online] Available at: https://www/foxgrp.com/healthcare-feasibility-studies/feasibility-study-of-a-hospital-project-5-phases/ [Accessed 10 June 2025]

29. Fatemeh Rangraz Jeddi, M. A. H., 2016. *National library of medicine.* [Online] Available at: https://pmc.ncbi.nlm.nih.gov/articles/PCM5074754/ [Accessed 10 June 2025]
30. Scheid, J. 2024. *Bright Hub pm.* [Online] Available at: https://www.brighthubpm.com/project-planning/72872-an-example-of-operational-feasibility/ [Accessed 12 June 2025].
31. Markham, S. (2025). Patient perspective on predictive models in healthcare: translation into practice, ethical implications and limitations? *BMJ Health & Care Informatics*, [online] 32(1), pp.e101153–e101153. doi:https://doi.org/10.1136/bmjhci-2024-101153.
32. Tshikomana, R.S. and Ramukumba, M.M. (2022). Implementation of mHealth applications in community-based health care: Insights from Ward-Based Outreach Teams in South Africa. *PLOS ONE*, 17(1), p.e0262842. doi:https://doi.org/10.1371/journal.pone.0262842.
33. www.csir.co.za. (n.d.). *Technology for better healthcare | CSIR*. [online] Available at: https://www.csir.co.za/technology-better-healthcare.
34. Knoesen, D. (2025). *BCX | AI and healthcare: paving the way for a healthier South Africa*. [online] BCX. Available at: https://www.bcx.co.za/industry-insights/ai-and-healthcare-paving-the-way-for-a-healthier-south-africa/.
35. www.heart.org. (n.d.). *How technology is improving health information access for the deaf community*. [online] Available at: https://www.heart.org/en/news/2021/07/12/how-technology-is-improving-health-information-access-for-the-deaf-community.
36. Bienefeld, N., Keller, E. and Grote, G. (2024). AI to Mitigate Healthcare Shortages: A Qualitative Analysis of Current and Future Work Conditions to Enhance Job Satisfaction and Wellbeing in Critical Care Nurses and Physicians (Preprint). *Journal of Medical Internet Research*, [online] 27, pp.e50852–e50852. doi:https://doi.org/10.2196/50852.
37. Sign Solutions (2023). *Overcoming communication barriers for Deaf patients*. [online] Sign Solutions. Available at: https://www.signsolutions.uk.com/overcoming-communication-barriers-for-deaf-patients-in-healthcare-settings/.
38. MichiganMedicine Builds Hearing Accommodations into the Patient Medical Record. (2024). *Default Digital Object Group*. doi:https://doi.org/10.1044/2024-1118-hearing-access-auds-slps.
39. McKee, M., James, T.G., Helm, K.V.T., Marzolf, B., Chung, D.H., Williams, J. and Zazove, P. (2022). Reframing Our Health Care System for Patients With Hearing Loss. *Journal of Speech, Language, and Hearing Research*, 65(10), pp.1–13. doi:https://doi.org/10.1044/2022_jslhr-22-00052.
40. Clear, S. 'N' (2023). *Improving Medical Care for Deaf and Hard of Hearing Patients*. [online] Safe'N'Clear, Inc. | The Communicator™ clear face mask. Available at: https://safenclear.com/improving-medical-care-for-deaf-and-hard-of-hearing-patients/.